Neural Surrogates for the Linear Stability of an Oscillatory Taylor–Couette Flow of an Upper-Convected Maxwell Fluid

A presentation of nine architectures

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Abstract

We present nine machine-learning surrogate models trained to predict the marginal stability curve $T_a(k, E)$ of an oscillatory Taylor–Couette flow between counter-oscillating cylinders for an Upper-Convected Maxwell (UCM) viscoelastic fluid. The ground truth is a Floquet eigenvalue problem with 720 stability branches indexed by elasticity number $E \in [10^{-4}, 10]$ and 23 functional descriptors. The architectures span sparse mixture-of-experts transformers, neural operators (FNO + latent diffusion), reinforcement-learning refiners (SAC, multi-agent SAC, diffusion-policy MARL), implicit neural representations (SIREN), branch-trunk operator learning (DeepONet), Chebyshev spectral regression, and a custom multi-mode envelope SIREN that encodes the Floquet mode-crossing structure. The envelope SIREN attains $\mathrm{MAE}_{\mathrm{phys}} = 0.549$ on the original input file, a $2.4\times$ improvement over the reference legacy pipeline.

1. Physical setting and Floquet eigenvalue problem

The flow is the perturbation of a 1D base profile $\boldsymbol{U}_0(r,t) = U_b(r)\cos(\omega t)\,\hat{\boldsymbol{e}}_\theta$ between two infinite concentric cylinders that counter-oscillate at angular frequency ω. Linear stability is governed by the Upper-Convected Maxwell constitutive equation,

$$\boldsymbol{\tau} + \lambda\Big(\partial_t\boldsymbol{\tau} + \boldsymbol{u}\cdot\nabla\boldsymbol{\tau} - (\nabla\boldsymbol{u})\boldsymbol{\tau} - \boldsymbol{\tau}(\nabla\boldsymbol{u})^\top\Big) = 2\mu\boldsymbol{D}(\boldsymbol{u}), \tag{1}$$

together with the momentum and incompressibility equations. After linearisation around $\boldsymbol{U}_0$, expansion in radial Chebyshev modes, and Floquet decomposition in time $\hat{\phi}(r,t) = e^{\sigma t}\sum_n \tilde{\phi}_n(r)\,e^{in\omega t}$, the stability problem becomes a generalised eigenvalue problem

$$\boldsymbol{A}(k, T_a, E)\,\hat{\boldsymbol{\phi}} \;=\; \sigma\,\boldsymbol{B}(k,E)\,\hat{\boldsymbol{\phi}}, \qquad \sigma = \sigma_r(T_a,k,E) + i\,\sigma_i. \tag{2}$$

The marginal stability curve is the locus $\sigma_r(T_a,k,E) = 0$ for the most unstable Floquet mode. Because several Floquet modes can compete, the curve has the lower-envelope structure

$$\boxed{T_a(k,E) \;=\; \min_{m=1,\dots,M} T_m(k,E)} \tag{3}$$

which produces piecewise-smooth cusps at parameter values where the active mode index switches.

2. Dataset

Source. The file `combined`$_d ata.csvcontains122\,820triples$`(T`$_a, k, E$) `obtained by sweeping the Floquet solver, yielding` 720 `branches segmented from` 420 `unique values of` E.

Branch descriptors. For each branch we extract a 23-dimensional feature vector $\boldsymbol{\xi} \in \mathbb{R}^{23}$ summarising the local geometry of the branch (widths, slopes, asymmetries, curvature at minimum, arc length, switch flags, etc.) and combine it with $\log_{10} E$ to obtain the context $\boldsymbol{\xi} \in \mathbb{R}^{24}$ supplied to every model.

Splits. Stratified 70/15/15 over $\log_{10} E$ with eight quantile bins gives 504/112/104 branches.

3. Architectures

3.1. Switch-Segment Sparse Transformer

A transformer with sparse Mixture-of-Experts blocks. For each input segment $\boldsymbol{h} \in \mathbb{R}^{T \times d}$ a router computes the top-k expert assignments

$$g_i(\boldsymbol{h}) = \mathrm{Softmax}_{\texttt{top-}k}(\boldsymbol{W}_g \boldsymbol{h})_i, \qquad \boldsymbol{h}' = \sum_{i=1}^{N} g_i(\boldsymbol{h})\, E_i(\boldsymbol{h}), \tag{4}$$

where $\{E_i\}$ are MLP experts. With $N = 8$ experts, $k = 2$ active per token, and 49M parameters, the model dominates the transformer benchmark while keeping inference cost bounded.

3.2. Fourier Neural Operator + Latent Diffusion

A 1D Fourier Neural Operator backbone implements the spectral block

$$\mathcal{F}^{-1}\big(\boldsymbol{R}_\theta \cdot \mathcal{F}(\boldsymbol{v})\big)(k) \;+\; \boldsymbol{W}\boldsymbol{v}(k), \tag{5}$$

where $\boldsymbol{R}_\theta$ is a learned complex multiplier truncated to the $M_F = 32$ lowest frequency modes, and $\boldsymbol{W}$ is a 1×1 convolution. A latent diffusion head with denoising score matching samples residual corrections via DDIM with $S = 50$ steps.

3.3. Single-Agent SAC Refiner (Conv1D + AttnPool)

A Soft Actor-Critic agent observes the surrogate baseline on a 49-pt window. The policy $\pi_\phi(a \mid s)$ is a tanh-squashed Gaussian, and learning maximises the entropy-regularised return

$$J(\phi) = \mathbb{E}_{(s,a)\sim\rho_\pi}\Big[\, Q_\theta(s,a) - \alpha \log \pi_\phi(a \mid s) \,\Big], \tag{6}$$

with the entropy temperature α adapted automatically. Features are extracted by a stack of 1D convolutions followed by attention pooling.

3.4. 3-SAC MARL with Cross-Attention

Three SAC agents specialise on *location*, *shape* and *geometry* of the correction. They share observations through a cross-attention block

$$\mathrm{CrossAttn}(\boldsymbol{Q}_i, \boldsymbol{K}, \boldsymbol{V}) = \mathrm{Softmax}\Big(\tfrac{\boldsymbol{Q}_i \boldsymbol{K}^\top}{\sqrt{d}}\Big)\boldsymbol{V} \tag{7}$$

which couples the agents while preserving heterogeneous policies.

3.5. Diffusion-Policy MARL (HYDRA, 7 agents)

A multi-agent system with 7 agents (manager, detector, asymmetry, geometry, shape, smoothness, validator). Each agent's policy is a denoising diffusion process

$$\boldsymbol{a}^{(t-1)} = \tfrac{1}{\sqrt{\alpha_t}}\Big(\boldsymbol{a}^{(t)} - \tfrac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\,\epsilon_\theta(\boldsymbol{a}^{(t)}, t, \boldsymbol{m})\Big) + \sigma_t \boldsymbol{z}_t, \tag{8}$$

conditioned on inter-agent messages $\boldsymbol{m}$ produced by a Graph Attention Network (GAT). Centralised critic + decentralised execution follows the MAPPO scheme; auxiliary algorithms include COMA, IQL, MADT and PBT.

3.6. SIREN with FiLM conditioning

Implicit neural representation [2] with sinusoidal activations and Feature-wise Linear Modulation (FiLM):

$$\boldsymbol{h}_{\ell+1} = \sin\big(\omega_0\,(\boldsymbol{W}_\ell \boldsymbol{h}_\ell + \boldsymbol{b}_\ell)\big), \qquad \boldsymbol{h}_\ell \leftarrow \boldsymbol{\gamma}_\ell(\boldsymbol{\xi}) \odot \boldsymbol{h}_\ell + \boldsymbol{\beta}_\ell(\boldsymbol{\xi}). \tag{9}$$

The choice $\omega_0 = 30$ matches Sitzmann *et al.*'s spectral prescription. With 0.83M parameters the model attains validation MAE 0.0098 on normalised T_a.

3.7. DeepONet with Fourier features

Operator learning [3]:

$$T_a(k) = \big\langle \mathrm{branch}(\boldsymbol{\xi}),\ \ \mathrm{trunk}(\gamma(k))\big\rangle + b_0, \tag{10}$$

with the Fourier embedding $\gamma(k) = \big[\,k,\ \{\sin(2\pi w_i k), \cos(2\pi w_i k)\}_{i=1}^{16}\,\big]$ where the angular frequencies w_i are log-spaced up to $\omega_{\max} = 16$.

3.8. Chebyshev Spectral Regressor

Each branch is decomposed on the truncated Chebyshev basis of the first kind:

$$T_a(k_{\mathrm{norm}}) = \sum_{n=0}^{N} c_n(\boldsymbol{\xi})\, T_n(k_{\mathrm{norm}}), \qquad N = 16, \tag{11}$$

where T_n satisfies $T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$. A 4-layer Transformer encoder maps $\boldsymbol{\xi}$ to the coefficients $\{c_n\}_{n=0}^{16}$. Although globally smooth, the truncated polynomial basis is plagued by Gibbs ringing at mode-crossing cusps.

3.9. Multi-Mode Envelope SIREN ★

Designed to honour the Floquet envelope structure of Eq. (3): $M = 4$ SIREN modes share a context encoder and the output is their hard-min envelope,

$$T_a(k, \boldsymbol{\xi}) \;=\; \min_{m=1,\dots,M} T_m(k, \boldsymbol{\xi}), \qquad T_m(k, \boldsymbol{\xi}) = f_m^{\mathrm{SIREN}}(k\,;\,\phi_{\mathrm{ctx}}(\boldsymbol{\xi})) + \beta_m, \tag{12}$$

with symmetry-breaking biases $\{\beta_m\}$. The hard min is piecewise-differentiable: gradients flow through the active mode only, so each T_m specialises on the parameter region where it is the lower threshold, exactly as a Floquet mode does.

4. Training: kink-weighted MSE and selective smoothing

Loss. Standard MSE under-weights the few samples near a cusp. We use a kink-weighted MSE

$$\mathcal{L}(\theta) = \frac{1}{BT} \sum_{b,t} w(t \mid T_a^{\texttt{true}}{}_b) \cdot \big(f_\theta(k_t, \boldsymbol{\xi}_b) - T_a^{\texttt{true}}{}_{b,t}\big)^2, \tag{13}$$

with the per-sample weight

$$w(t \mid T_a^{\texttt{true}}) = 1 + (\alpha - 1)\, \sigma\big(2(z_t - \tau)\big), \qquad z_t = \frac{|\,\Delta^2 T_a^{\texttt{true}}{}_t\,|}{\mathrm{std}_t\, |\Delta^2 T_a^{\texttt{true}}{}_t|}, \tag{14}$$

where Δ^2 is the discrete second difference, σ is the sigmoid, $\alpha = 5$, and $\tau = 2$. Points whose second difference exceeds 2σ are weighted up to $5\times$, focusing the gradient on the cusps.

Selective smoothing (post-processing). For visualisation, predictions are smoothed by a Savitzky-Golay filter outside the kink zones. Defining the soft mask $M(t) = \mathrm{box}_\nu\big(\sum_{k\in\mathcal{K}} \mathbb{1}_{[\,k-e,\,k+e\,]}(t)\big)$ on the kink set $\mathcal{K}$ with edge e, the displayed curve is

$$\widetilde{T_a}(t) = M(t)\, T_a(t) + (1 - M(t))\, \big(\mathrm{savgol}\, T_a\big)(t). \tag{15}$$

The mask preserves cusps exactly while flattening parasites in smooth regions.

5. Results

Evaluation on the 720 branches of combined_data.csv in physical units of T_a:

Model	MAE	RMSE	Kink-zone MAE	MaxErr
Multi-Mode Envelope SIREN ★	**0.549**	1.10	**0.785**	26.7
SIREN + FiLM	0.897	1.55	1.30	30.1
DeepONet	1.34	2.52	2.10	48.0
Sparse-MoE-T + 3-SAC (legacy ref.)	1.31	-	-	-
Chebyshev Spectral	2.53	7.27	3.33	182.9

The custom envelope architecture beats every smooth baseline and the legacy multi-agent pipeline by a factor of ≈ 2.4, while having the lowest error on kink zones and the lowest tail (MaxErr).

6. Conclusion

Encoding the Floquet mode-crossing structure $T_a = \min_m T_m$ as an architectural inductive bias, combined with a kink-weighted MSE loss, yields a surrogate that fits marginal stability curves with sub-percent accuracy on 720 branches. The same recipe applies to other eigenvalue stability problems where the threshold curve is a lower envelope of competing modes.

References

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